CORA: RANDOMIZED DECODING FOR COMPRESSED UPDATE INVERSION AND ITS RECONSTRUCTION LIMITS ACROSS BIT WIDTHS

Anonymous Author(s)

Anonymous Affiliation

ABSTRACT

Communication compression leaves a federated client's update numerically ambiguous: the server observes integer codes and a scale, not the original values, and naive dequantization commits to one completion, the cell center. We ask whether searching the other admissible completions helps an attacker, and how that changes as the cells widen. We present CoRA (Consistency-Ranked Attack), which samples updates reproducing the received codes and scale exactly, reconstructs one image candidate from each through a CPA-based pipeline, and returns the candidate ranked best by a score computed from public information alone. Across per-tensor sweeps from 8 to 4 bits on ImageNet and CIFAR-100, perceptual quality degrades monotonically with precision while PSNR and SSIM recover at low precision as reconstructions flatten. Randomized decoding improves LPIPS in eight of ten quantization conditions; on CIFAR-100 its margin is largest at 4 bits, whereas on ImageNet at 4 bits a better draw existed among the eight and the ranking did not return it. The binding limit there is selection, not sampling.

***Index Terms*—** Update inversion, randomized decoding, quantization, source separation, reconstruction quality.

1. INTRODUCTION

A client's model update is a function of its private training data, and attacks that invert this mapping recover the inputs themselves. Optimization-based attacks match synthetic gradients to the observed ones [1, 2]; the Cocktail Party Attack (CPA) instead treats a fully connected gradient as a linear mixture of input features and unmixes it by independent component analysis [3]. Both read the observed update as a numerical quantity. Under communication compression there is none to read: the server receives integer codes and a scale, and must first decide what update those codes stand for. That decision is normally settled by convention. Naive dequantization maps each code to its cell center, one element of the set of updates consistent with the observation, chosen because it is deterministic and cheap. Every other element explains the same packet equally well, and sending a different one through the same pipeline yields a different image. Two questions follow: does searching these alternatives recover more of the private input than the center does, and how does the answer change as coarser quantization widens the cells?

We study both with CoRA (Consistency-Ranked Attack), which draws updates reproducing the received codes and scale exactly, separates source features from each draw, inverts them into one image candidate per draw, and ranks the candidates by a score built solely from the public model and the decoded sources. It returns a single reconstruction rather than an average, and no private data enters the search or the selection. Across per-tensor sweeps from 8 to 4 bits on ImageNet and CIFAR-100, quality falls steadily with precision and randomized decoding does not reverse that trend. It does shift it: LPIPS improves in eight of the ten quantization conditions, and on CIFAR-100 the margin grows as the cells widen. On ImageNet at 4 bits the margin is negative, yet one of the eight draws came out ahead of the naive control and the ranking did not select it. We therefore report where candidate search pays, and where its limit lies in selection rather than sampling.

2. OBSERVATION MODEL

2.1. Client update and attacker knowledge

We consider an attacker observing one client's local update in a federated round, with no secure aggregation in place. Starting from public weights, the client runs T SGD steps on the same batch of B images, with the convolutional extractor frozen and the fully connected layers trained. The attacker knows the starting model, the batch size and the local training configuration, and receives the quantized update of every fully connected tensor with its scale. The private images, their labels, the features they induce and the uncompressed update reach neither the attack nor its selector.

Let $\mathbf{H} = [\phi(x_1), \dots, \phi(x_B)]^\top$ collect the frozen features entering the first fully connected layer. Writing the gradient at step t as $\mathbf{A}_t\mathbf{H}$, with loss averaging absorbed into $\mathbf{A}_t$, that layer's update is

$$\mathbf{U} = \mathbf{W}^{(T)} - \mathbf{W}^{(0)} = -\eta\Big(\sum_{t=0}^{T-1} \mathbf{A}_t\Big)\mathbf{H}. \tag{1}$$

Because the extractor is frozen and every step reuses the same batch, $\mathbf{H}$ does not depend on t: the whole T-step update stays

a linear mixture of the same B feature vectors, its rows lying in their span so that $\operatorname{rank} \mathbf{U} \leq B$. This is the structure CPA exploits, and why we attack the first fully connected weight matrix; at later layers the inputs change between steps and Eq. (1) no longer holds.

2.2. Signed per-tensor quantization

Each tensor is quantized symmetrically under a single scale,

$$\begin{aligned} Q_b &= 2^{b-1} - 1, \qquad & s &= \frac{\|\mathbf{U}\|_\infty}{Q_b}, \\ q_j &= \operatorname{round}_{\text{even}}(U_j/s), \quad & q_j &\in \{-Q_b, \ldots, Q_b\}, \end{aligned} \tag{2}$$

with ties broken to even so that encoding is deterministic. The attacker receives the codes q and the FP32 scale s, never $\mathbf{U}$. We treat b as a numerical precision: the codes are not bit-packed in our artifacts, so we make no claim about realized bandwidth.

A code names a cell, not a value. Let $\mathcal{C}(q, s)$ be the set of updates the encoder maps to exactly this packet,

$$\begin{aligned} \mathcal{C}(q, s) = \big\{ \mathbf{U}' : \operatorname{round}_{\text{even}}(U'_j/s) = q_j \ \forall j, \\ \|\mathbf{U}'\|_\infty / Q_b = s \big\}. \end{aligned} \tag{3}$$

The first condition confines each coordinate to an interval, in exact arithmetic $[s(q_j - \frac{1}{2}),\ s(q_j + \frac{1}{2})]$ for interior codes and bounded for extremal ones. The second couples the coordinates: since s derives from $\|\mathbf{U}\|_\infty$, a candidate reproducing every code but altering the largest magnitude would imply a different scale, hence a different packet. $\mathcal{C}(q, s)$ is therefore not a product of intervals, and any sampler must respect that coupling. Naive decoding returns the cell centers, $U_j^{\text{ctr}} = sq_j$, one member of $\mathcal{C}(q, s)$ among many.

3. CORA: RANDOMIZED DECODING AND RANKING

3.1. Sampling compatible numerical updates

From the received packet CoRA draws K candidate first-FC matrices $\widetilde{\mathbf{U}}^{(m)} \in \mathcal{C}(q, s)$. Every coordinate is sampled uniformly between the inclusive representable FP32 endpoints of its own cell, and one extremal coordinate is pinned to the largest magnitude the scale admits. Since no cell reaches past that magnitude, the draw satisfies both conditions of Eq. (3) by construction; we re-encode each draw and require an exact match to the packet, as a check rather than a filter. These draws are not attempts at accuracy: a sample is no closer to $\mathbf{U}$ than the center is. What differs is where inside the cell set each draw lands, and because everything downstream is nonlinear, draws indistinguishable at the packet level can separate into visibly different images.

3.2. Source separation and feature inversion

We separate each decoded matrix by independent component analysis, using either the original CPA optimizer or a FastICA initialization with a short gradient refinement; Section 4.1 states which applies where. ICA recovers sources only up to permutation and sign, so we align each candidate's source rows to those of the center decoding by Hungarian assignment on absolute cosine similarity. Alignment permutes and flips rows and nothing else, leaving norms untouched and never averaging sources across candidates. Because the true features are post-ReLU and hence nonnegative, we take magnitudes $z_{m,i} = |\hat{h}_{m,i}|$ as inversion targets, and feature inversion optimizes images $\hat{\mathbf{X}}_m$ so that $\phi(\hat{x}_{m,i})$ matches $z_{m,i}$ through the public extractor.

3.3. Public screening and continuation

After a short inversion budget we score each candidate by

$$J_m = \frac{1}{B} \sum_{i=1}^{B} [1 - \cos(\phi(\hat{x}_{m,i}), z_{m,i})] + 0.1\, \mathrm{TV}_{\text{norm}}(\hat{\mathbf{X}}_m), \tag{4}$$

evaluated on a copy of the images clamped to the valid pixel range in the model's normalized input space. Every term is available to the attacker. Consistency in CoRA therefore means feature agreement regularized by total variation, not a test of whether the reconstruction would reproduce the packet if retrained and recompressed; Section 5 returns to that gap.

Ranking happens once, at screening, and is never revisited. With $K = 8$, each candidate receives 5,000 inversion iterations; the three smallest J_m continue to 25,000 with optimizer and scheduler state preserved, ties breaking on ascending member index; and the candidate ranked first at 5,000 iterations is the reported output. Continuing three rather than one leaves the runner-up endpoints available for Section 4.4. No private quantity re-orders the candidates, and no reconstructions are averaged. The naive control decodes to cell centers and runs the same backend for 25,000 iterations, so the center decoding serves twice, as this control and as the alignment reference above.

4. EXPERIMENTAL EVALUATION

4.1. Settings and evaluation protocol

We attack pretrained VGG16 on one fixed ImageNet batch of $B = 16$ images, and **[TBD: CIFAR-100 model]** on CIFAR-100 over three independent batches of $B = 8$, holding the victim data and local-training configuration constant across compression levels within each dataset. The client uses $T = 10$ and SGD at learning rate 10^{-3} with zero momentum, zero weight decay and dropout disabled; only fully connected parameters are trained, and every tensor is quantized under a

single scalar scale. We ask for B sources, the bound implied by Eq. (1). Separation uses the original CPA optimizer for 25,000 iterations everywhere except ImageNet quantization, where FastICA runs to a cap of 200 iterations at tolerance 10^{-6} followed by 4,000 refinement steps at learning rate 10^{-3}. Randomized decoding draws eight candidates, on ImageNet with seeds $1234 + 1000003m$ for $m = 0, \ldots, 7$ and on CIFAR-100 with **[TBD: CIFAR-100 seeds]**. Feature inversion uses Adam at initial learning rate 0.1 under a cosine schedule, with feature and total-variation weights 1 and 0.1.

We report mean PSNR, SSIM and VGG-based LPIPS [4]. Nothing private is touched until public selection is fixed. Because source separation is permutation-ambiguous, the evaluator then computes one Hungarian assignment per candidate and reuses it for all three metrics and every displayed reconstruction; the assignment belongs to evaluation and is never visible to the attack. Its cost is pixel MSE for ImageNet quantization, and one minus the absolute cosine between true and recovered source features for ImageNet sparsification and for CIFAR-100. All reconstructions enter the reported means.

4.2. Results across compression levels

Scores in Table 1 compare across datasets only within a Naive/CoRA pair, since the two settings differ in model, image size and batch size at once. Within CIFAR-100 the panels are directly comparable, sharing separator, batches and evaluation, so the two codecs can be read against each other there. The ImageNet panels cannot: quantization uses a different separator and matching, and the roughly 10 dB PSNR gap between them reflects that rather than codec strength.

Quantization degrades perceptual quality monotonically on both datasets. From 8 to 4 bits, naive LPIPS rises from 0.559 to 0.807 on ImageNet and from 0.561 to 0.731 on CIFAR-100, with CoRA tracking the same direction. PSNR and SSIM do not follow. Both fall through the middle of the sweep and recover at 4 bits — on CIFAR-100 from 10.22 dB at 6 bits back to 11.27 dB, on ImageNet with SSIM returning to its 8-bit value — while LPIPS keeps worsening. We read the divergence as the familiar one: a reconstruction collapsing toward a flat, low-contrast field is rewarded by pixel- and structure-level measures and penalized by a perceptual one, and Fig. 1 shows that flattening directly. We therefore take LPIPS as the primary quantity.

CIFAR-100 supplies a reference the quantization sweep otherwise lacks. At 50% retention in panel (b) the discarded entries are already zero, so that condition transmits the first-FC update intact under the same backend, and its LPIPS of 0.488 is this pipeline's ceiling. Against it, 8-bit quantization costs 0.073 and 4-bit costs 0.243, while discarding nine entries in ten costs 0.082. Keeping one entry in ten therefore leaks about as much as quantizing every entry to 8 bits, and considerably more than quantizing to four.

On LPIPS, randomized decoding helps in eight of the ten quantization conditions and three of the four non-trivial sparsification conditions. The ImageNet margin is small and not ordered in precision, with reversals at 7 and 4 bits. On CIFAR-100 it is positive at every bit width and widens as compression grows, from 0.008 at 8 bits to 0.047 at 4 bits; pooled over all CIFAR-100 conditions and images, CoRA leads on all three metrics, at 0.582 against 0.596 in LPIPS. We note the contrast between datasets without resolving it: they differ in model, image size, batch size, separator and matching simultaneously.

Table 1. Naive versus randomized decoding (PSNR↑, SSIM↑, LPIPS↓). Dashes denote runs not available. Every block uses the original CPA separator except ImageNet quantization, which uses FastICA initialization with gradient refinement. ImageNet rows cover one batch of 16 images; CIFAR-100 rows pool three batches of 8. The two CIFAR-100 panels are therefore directly comparable; the two ImageNet panels are not.

(a) Quantization

Dataset	Method	Metric	8-bit	7-bit	6-bit	5-bit	4-bit
CIFAR-100	Naive	PSNR	11.49	11.07	10.22	10.65	11.27
		SSIM	0.166	0.134	0.062	0.061	0.075
		LPIPS	0.561	0.605	0.685	0.706	0.731
	CoRA	PSNR	11.70	11.33	10.44	10.62	10.70
		SSIM	0.167	0.154	0.080	0.061	0.057
		LPIPS	0.553	0.585	0.673	0.681	0.684
ImageNet	Naive	PSNR	12.06	11.80	11.64	11.55	11.52
		SSIM	0.292	0.276	0.260	0.262	0.292
		LPIPS	0.559	0.612	0.696	0.783	0.807
	CoRA	PSNR	12.06	11.83	11.67	11.55	11.50
		SSIM	0.307	0.283	0.269	0.256	0.258
		LPIPS	0.556	0.633	0.666	0.749	0.808

(b) Sparsification: retained fraction of FC1

Dataset	Method	Metric	50%	30%	20%	10%
CIFAR-100	Naive	PSNR	13.70	13.45	13.21	11.29
		SSIM	0.265	0.248	0.252	0.164
		LPIPS	0.488	0.510	0.510	0.570
	CoRA	PSNR	13.70	13.47	13.10	11.64
		SSIM	0.265	0.257	0.252	0.166
		LPIPS	0.488	0.489	0.522	0.561
ImageNet	Naive	PSNR	22.29	22.14	21.87	21.53
		SSIM	0.334	0.314	0.290	0.263
		LPIPS	0.486	0.509	0.553	0.632
	Random fill	PSNR	—	—	—	21.51
		SSIM	—	—	—	0.261
		LPIPS	—	—	—	0.630

4.3. Sparsification

The first-FC update keeps its largest-magnitude entries at retained fractions of 50, 30, 20 and 10%, and naive decoding zero-fills the rest. On CIFAR-100 these runs share the sep-

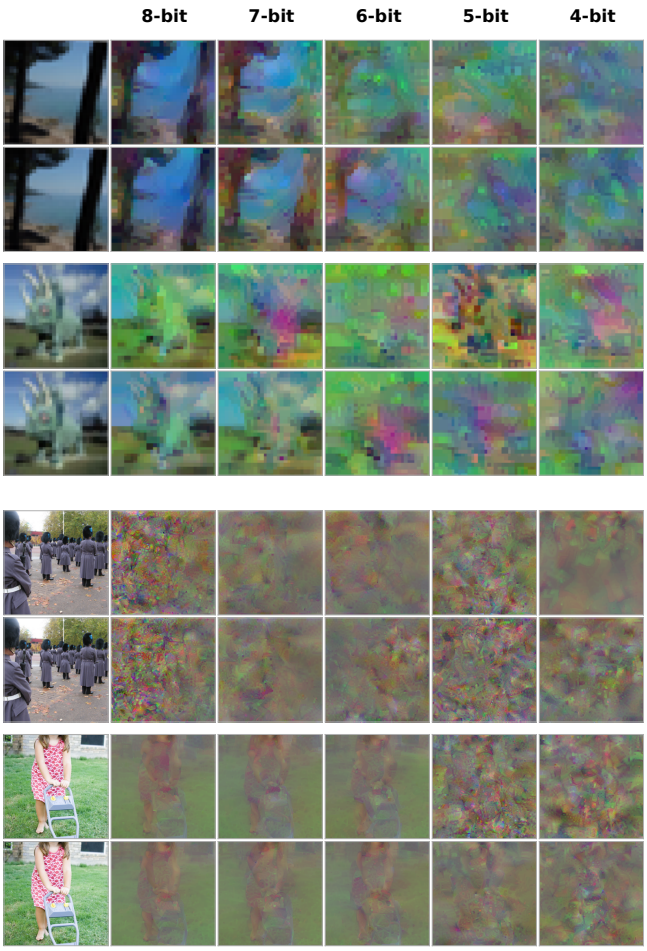


Fig. 1. Reconstructions at each bit width. Within every block the left column is naive center decoding and the right the publicly selected CoRA output, both at 25,000 inversion iterations; the leftmost column is the private image. Upper two rows are CIFAR-100 registered indices 11 and 24, lower two ImageNet indices 0 and 1, all fixed across conditions, with saved Hungarian assignments reused unchanged. At 4 bits the naive reconstructions flatten while CoRA keeps more high-frequency texture — the behaviour PSNR and SSIM reward and LPIPS penalizes.

arator, batches and evaluation of the quantization sweep; the ImageNet study is separate, using the original CPA optimizer with recovered-source cosine matching. At 50% retention the discarded entries are already zero on both datasets: by Eq. (1) the update inherits the sparsity of the post-ReLU features, so more than half its coordinates vanish before any sparsification. The fill radius τ is then zero, random filling reduces to zero filling, and the two CIFAR-100 rows coincide exactly, as they must. Below that level we fill discarded entries uniformly from $[-\tau, \tau]$ for τ the smallest retained magnitude. On CIFAR-100, CoRA improves LPIPS at 30% (0.510 to 0.489) and 10% (0.570 to 0.561) and loses at 20%. The ImageNet run at 10% instead ranks twenty candidates by an update-consistency score, which replays the known local SGD with public-model predicted labels, recompresses, and measures the relative ℓ_2 distance from the packet; selection is applied twice rather than fixed at screening. It yields LPIPS 0.630 against 0.632 for naive decoding, better on one metric and worse on two.

4.4. What the selector finds, and what it misses

At 4 bits on ImageNet the first draw, member 00, reached PSNR 11.56 dB, SSIM 0.261 and LPIPS 0.806, marginally ahead of the naive control on PSNR and LPIPS. The selector ranked member 06 first instead, and member 06 is worse than naive on all three metrics. The margins are small, but the ordering is not: among the draws whose endpoints we can compare, the ranking did not return the better one. That row therefore reflects selection at least as much as sampling, while the corresponding CIFAR-100 row shows the same sampler at its largest margin of the sweep. Member 00 reached the full budget only because it ranked third, not as a separately evaluated $K = 1$ attack, so this is an existence observation rather than a measured property of the selector. We nonetheless report the publicly selected candidate throughout.

4.5. Budget and scope

Randomized search costs $8 \times 5{,}000 + 3 \times 20{,}000 = 100{,}000$ inversion iterations per condition, plus 25,000 for the naive control and one separation per draw and for the center reference. Final per-candidate budgets match while total search budgets do not, and we make no compute-matched claim: the cosine schedule is fixed at a 25,000-iteration horizon, so granting the control the same total would require a different schedule, not a longer run. Every ImageNet number comes from a single victim batch; CIFAR-100 numbers pool three independent batches of eight images, which we do not break out here. Several differences between methods are of the same order as effects we have not bounded.

5. DISCUSSION

Degradation with precision needs no appeal to our sampler: naive decoding degrades too, and the packet carries less about $\mathbf{U}$. What the sampler adds is spread. Cell width is $s = \|\mathbf{U}\|_\infty / Q_b$, so within a dataset it scales as $1/Q_b$ and grows by $127/7 \approx 18$ between 8 and 4 bits. Wider cells cut both ways, leaving more room for a draw to land nearer $\mathbf{U}$ than the center and more room for it to land further away, and the ranking decides between them. On CIFAR-100 the margin is largest at 4 bits, where the cells are widest; on ImageNet at 4 bits it is negative, yet Section 4.4 exhibits a draw that came out ahead of the center. That row measures selection,

not sampling. We offer this as the reading most consistent with the data, not as a mechanism the sweep isolates.

Ranking is limited in a way we can state precisely. The targets in Eq. (4) are estimated from the compressed update itself, so a candidate can agree with them closely and resemble the private image poorly. Nor does exact re-encoding establish the stronger property one would want: that local training on the reconstructed images would reproduce the packet. The ImageNet sparsification study ranks by exactly that test, but on a different codec and backend, so these results cannot say which selector is better. Bounding what ranking costs would require continuing every draw to the full budget.

6. RELATED WORK

Optimization-based gradient inversion reconstructs inputs by matching observed and synthetic gradients [1, 2]. CPA instead treats fully connected gradients as blind source separation solved by ICA, scaling to larger batches [3], and Zhang et al. [5] survey the resulting landscape. On compression as a defense, Yue et al. [6] show image information can survive quantization, sparsification or perturbation; Li et al. [7] reconstruct under gradient degradation with a GAN prior learned from public images; and Ding et al. [8] target sparsified gradients by compressing synthetic gradients during reconstruction. CoRA instead studies randomized decoding and public candidate selection without a pretrained prior: the sender's quantization is deterministic, and the attacker samples alternative numerical updates compatible with the observation, ranks them by feature agreement and total variation, and reports one selected reconstruction.

7. CONCLUSION

CoRA reconstructs quantized local updates by sampling compatible decodings and ranking the resulting candidates with a public feature-based score. Per-tensor bit sweeps on ImageNet and CIFAR-100 show LPIPS worsening monotonically from 8 to 4 bits, while PSNR and SSIM recover at low precision as the reconstructions flatten. Randomized decoding improves LPIPS in eight of ten conditions; its margin at 4 bits is the largest we measured on one dataset and negative on the other, where a better draw existed and the ranking did not return it. The measured curves and reconstructions describe where the procedure retains image information and where its limitations become pronounced.

8. REFERENCES

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